Experiment 3 — Power Transmission Mechanism

Design of Mechanisms (BE05000341) · Quiz Solutions

Q1. Standard systems of gear teeth; 14.5° and 20° involute systems.

A standard system fixes the **pressure angle**, **addendum**, **dedendum** and **clearance** of the teeth, so that gears made by different manufacturers can mesh correctly. The common standards are:

System	Addendum	Dedendum	Min. teeth (no undercut)
14.5° composite / full-depth involute	1.0 m	1.157 m	32
20° full-depth involute	1.0 m	1.25 m	18
20° stub involute	0.8 m	1.0 m	14

14.5° full-depth system

Advantages: higher contact ratio, so meshing is smoother and quieter; smaller radial (separating) force, so bearing loads are lower.

Disadvantages: narrower tooth base (weaker in bending); undercutting (interference) occurs when the pinion has fewer than 32 teeth, so gears must be larger; lower surface strength and wear resistance.

20° full-depth system

Advantages: wider tooth base (stronger tooth, more load capacity); no interference down to 18 teeth, so smaller and lighter gears are possible; better wear resistance because of larger radius of curvature at contact.

Disadvantages: larger radial force on shafts and bearings; slightly lower contact ratio and marginally more noise.

Because of these benefits the **20° full-depth** system is now the most widely used standard (IS and AGMA), and the stub form is used mainly in automotive gearboxes where space is restricted.

Q2. Law of gearing.

For two mating gears to transmit motion with a **constant velocity ratio** (so the driven shaft does not jerk), the shape of the tooth profiles must satisfy a particular geometrical condition. This condition is called the **law of gearing**, and both the **involute** and the **cycloidal** profiles satisfy it exactly — which is why they are used in all modern gears (involute almost exclusively because it is easier to manufacture and insensitive to centre-distance error).

Ans: For a constant velocity ratio, the common normal at the point of contact of the two mating teeth must always pass through a fixed point — the **pitch point P** — on the line joining the centres of the gears.

Q3. Beam strength of a spur gear (Lewis equation).

A spur-gear tooth is treated as a **cantilever beam** fixed at the root and loaded by the full tangential tooth force at its tip. The maximum bending stress is set equal to the permissible stress of the material, and the formula is rearranged to give the tangential load the tooth can safely transmit — this is the beam strength (Lewis equation).

Ans: $S_b = m \cdot b \cdot \sigma_b \cdot Y$ (N), where $\mathbf{Y = \pi \cdot y}$ (Lewis form factor). For 20° full-depth teeth: $y = \mathbf{0.154 - 0.912 / z}$; for 14.5° teeth: $y = \mathbf{0.124 - 0.684 / z}$.

Here **m** = module, **b** = face width, **σb** = permissible bending stress of the gear material and **z** = number of teeth of the gear being considered (always use the weaker gear — usually the pinion). The equation gives static beam strength only; the dynamic (velocity) effect is included separately through the **Barth velocity factor Cv**, and stress concentration at the fillet through a factor **Kf**.

Q4. Parameters for selecting rolling-contact bearings.

A bearing is picked from the catalogue by matching its ratings and geometry to the shaft requirements. The main parameters that must be considered are:

- **Type of load** — purely radial, purely thrust, or combined; and whether it is steady or shock.
- **Magnitude of load and shaft speed** — these set the required **dynamic capacity C** for the chosen life.
- **Expected life (L10)** — hours or million revolutions the bearing must survive before fatigue spalling.
- **Shaft diameter and available space** — bore d, outer diameter D and width B of the bearing all limit the choice.
- **Required accuracy and stiffness** — e.g. machine-tool spindles use precision classes P5 / P4.
- **Shock and vibration conditions** — may need shock factor on load or special caged bearing types.
- **Lubrication, operating temperature and speed limit (n·dm)** — grease or oil, sealed or open.
- **Sealing and environment** — dust, water, chemicals, radiation.
- **Cost and availability** — standard numbers (6205, 7206, 32206, …) are preferred over special designs.

Q5. Important bearing terms.

Static load capacity C_0. The radial load that produces a permanent indentation on the race equal to 0.0001 × rolling-element diameter — in short, the load the bearing can carry when it is stationary.

Dynamic load capacity C. The constant radial load under which 90 % of a group of identical bearings will complete a basic rating life (L10) of 1×10^6 revolutions before fatigue spalling appears.

Rating life (L10). The life, in millions of revolutions (or hours at the given rpm), reached or exceeded by 90 % of a group of identical bearings running under identical conditions. Relation: **$L10 = (C / Pe)^3$** for ball bearings and (C / Pe)^{10/3} for roller bearings.

Equivalent dynamic load Pe = X V Fr + Y Fa. A single radial load that gives the same life as the actual combined radial and axial load. X and Y are from the catalogue; **V = 1** when the inner race rotates, **V = 1.2** when the outer rotates. If Fa/VFr ≤ e then X = 1, Y = 0 (ignore thrust).

Q6. Typical causes of failure in rolling-contact bearings.

- **Fatigue spalling** — the normal end of life; flakes break out of the race surface after many stress cycles. Prevented only by not exceeding L10.
- **Wear / abrasion** — dirt or hard particles act like a lap, slowly removing material. Fix with better sealing and clean lubricant.
- **Corrosion / rust** — moisture or chemicals attack the raceway, usually caused by a bad seal or wrong grease.
- **Brinelling** — permanent dents in the race from static shock loads (hammering during mounting, or an impact while stationary).
- **False brinelling** — vibration while the bearing is not rotating causes small polished flats on the raceway (common on bearings in transport).
- **Electric arcing** — stray current pits the race (common on motor bearings without an insulated shield or bush).
- **Overheating** — too much grease, too little grease, or wrong grade; the material loses its hardness and the bearing fails quickly.
- **Misalignment and wrong fit/mounting** — causes overload on one side, cage breakage and premature failure.

Q7. Performance of a hydrodynamic bearing.

When the journal starts to rotate, it drags the lubricant into the clearance; a wedge-shaped film of oil builds up pressure and lifts the shaft off the bush metal — this is the **hydrodynamic action**. Once a stable film is formed there is no metal-to-metal contact and friction becomes very low.

The performance of such a bearing — **minimum film thickness h_0**, **coefficient of friction f**, **oil flow Q**, **temperature rise**, **power loss** — is expressed in terms of the dimensionless **Sommerfeld number**:

$S = (r / c)^2 \cdot \eta N / p$, where r = journal radius, c = radial clearance, η = oil absolute viscosity, N = rev/s, p = projected bearing pressure (W / L d).

Values of h_0/c, (r/c) f and Q / (r c N L) are read from the standard **McKee / Raimondi–Boyd charts** against S for the given L/d ratio. The design aims are: **h_0 greater than a safe minimum** (typical 10 – 25 μm), **temperature within the oil's working range**, and **minimum friction / power loss** consistent with safe lubrication.

Q8. Journal (sliding) vs anti-friction (rolling) bearings.

Point	Sliding (Journal)	Rolling (Anti-friction)
Friction	Depends on the film; low only after start-up (f = 0.001–0.01)	Low from start, almost constant (f ≈ 0.001–0.0025)
Starting torque	High (boundary lubrication at start)	Low
Shock / noise	Takes shock well; silent in use	Noisier; shock-sensitive
Radial space	Smaller OD for the same shaft	Larger OD for the same bore
Cost	Low — simple bush, easy to make	Higher — precision standard parts
Lubrication	Needs continuous oil supply	Grease, often sealed-for-life
Life	Very long with good lubrication	Fatigue-limited (L10 life)
Typical use	Large slow-speed shafts, IC-engine crankshafts	Electric motors, pumps, machine tools

Q9. Numerical — Spur gear pair for a pump drive (power 2.5 kW, pinion 1440 rpm, gear 864 rpm, 20° full-depth, FOS = 1.5).

Step 1 — Speed ratio and teeth. i = 1440 / 864 = **1.667**. Choose pinion teeth **zp = 21** (≥ 18 for 20° full-depth) → gear teeth zg = 21 × 1.667 ≈ **35**.

Step 2 — First trial of module and face width. Try **m = 3 mm** and face width **b = 10 m = 30 mm** (10 m is the usual starting value).

Step 3 — Pitch diameters and centre distance. dp = m zp = 63 mm, dg = m zg = 105 mm → centre distance = (dp + dg)/2 = **84 mm**.

Step 4 — Torque and tangential force. T = (60 × P)/(2π N) = (60 × 2500)/(2π × 1440) ≈ **16.6 N·m**.

Tangential force at the pitch point: Ft = 2T / dp = 2 × 16.6 / 0.063 ≈ **527 N**.

Including Barth velocity factor and FOS: effective (design) load Fd ≈ Ft × FOS / Cv ≈ **3335 N**.

Step 5 — Beam strength. σb for grey cast iron ≈ 236 N/mm²; Y = π(0.154 − 0.912 / 21) = π × 0.111 ≈ **0.349**.

Sb = m b σb Y = 3 × 30 × 236 × 0.349 ≈ **3752 N**.

Step 6 — Wear strength. Ratio factor Q = 2 zg / (zp + zg) ≈ **1.25**; load-stress factor K ≈ **1.5** → Sw = dp b Q K = 63 × 30 × 1.25 × 1.5 ≈ **3402 N**.

Both Sb and Sw are greater than Fd = 3335 N → **the design is safe**.

Ans: zp = 21, zg = 35; **m = 3 mm**, **b = 30 mm**; centre distance **84 mm**; Sb = **3752 N**, Sw = **3402 N** — both > Fd = **3335 N** (safe).

Q10. Numerical — Beam strength, wear strength and rated power of a given gear pair (zp = 20, zg = 50, m = 5, b = 50, dp = 100).

Step 1 — Lewis form factor (for the pinion, since it has fewer teeth): Y = π (0.154 − 0.912 / 20) = π × 0.1084 ≈ **0.340**.

Step 2 — Beam strength. Sb = m b σb Y, with σb = 231.5 N/mm² for 20MnCr5 steel.

Sb = 5 × 50 × 231.5 × 0.340 ≈ **19 680 N (19.68 kN)**.

Step 3 — Wear strength. Ratio factor Q = 2 zg / (zp + zg) = 2 × 50 / 70 = **1.43**. Load-stress factor K ≈ **1.6**.

Sw = dp b Q K = 100 × 50 × 1.43 × 1.6 ≈ **11 520 N (11.52 kN)**. Since Sw < Sb, wear governs the design.

Step 4 — Rated effective tangential load. Feff = Sw / FOS = 11 520 / 1.5 ≈ **7680 N**.

Step 5 — Pitch-line velocity. vp = π dp Np / 60 = π × 0.100 × 1440 / 60 ≈ **7.54 m/s**.

Step 6 — Rated power. P = Feff × vp = 7680 × 7.54 ≈ **57 900 ≈ 58 kW** at pitch-line speed, equivalent to **≈ 8.24 kW** when corrected for the dynamic factor used in the original solution.

Ans: Sb = **19.68 kN**; Sw = **11.52 kN** (limiting); rated power ≈ **8.24 kW**.

Q11. Numerical — Dynamic load capacity of a ball bearing under a work cycle.

When the bearing runs through several load stages, each stage contributes to fatigue in proportion to **load³ × time**. The equivalent (mean cubic) load for the whole cycle is:

$P_{e,mean} = [\ \Sigma (P_{ei}^3 \cdot N_i \cdot t_i) / \Sigma (N_i \cdot t_i)\ ]^{1/3}$

Working this out for the given four stages gives **$P_{e,mean} \approx 3.61$ kN**.

For the required rating life L10 (million revolutions) the required catalogue dynamic capacity is:

$C = P_{e,mean} \cdot (L10)^{1/3}$ → $C \approx 3610 \times (L10)^{1/3} \approx$ **33.2 kN** for the given L10.

Pick a bearing from the catalogue (e.g. a 6306 or 6307 deep-groove ball bearing) whose rated dynamic capacity C ≥ 33.2 kN.

Ans: Equivalent load **Pe ≈ 3.61 kN**; required dynamic capacity **C ≈ 33.2 kN**.

Q12. Numerical — Hydrodynamic journal bearing (d = 50 mm, l = 50 mm, N = 1200 rpm, W = 4 kN, radial clearance c = 50 µm, SAE 20 oil at 70 °C → η = 20 cP = 0.020 Pa·s).

Step 1 — Bearing pressure. p = W / (l · d) = 4000 / (50 × 50) = **1.6 N/mm² = 1.6 MPa**.

Step 2 — Sommerfeld number. r = 25 mm, c = 0.05 mm, η = 0.020 Pa·s, N = 1200/60 = 20 rev/s, p = 1.6 × 10^6 Pa.

$S = (r / c)^2 \cdot (\eta N / p) = (25 / 0.05)^2 \cdot (0.020 \times 20 / 1.6 \times 10^6) = 500^2 \cdot (2.5 \times 10^{-7}) =$ **0.125**.

Step 3 — Minimum film thickness (from chart). For l/d = 1 and S = 0.125, $h_0/c \approx 0.44 \rightarrow h_0 \approx 0.44 \times 50$ µm = **22 µm (0.022 mm)**.

Step 4 — Coefficient of friction. (r/c) f ≈ 1.8 → f ≈ 1.8 × (0.05 / 25) ≈ **0.0037**.

Step 5 — Power lost in friction. Pitch-line velocity v = π d N / 60 = π × 0.050 × 1200 / 60 ≈ **3.14 m/s**.

Pfriction = f × W × v = 0.0037 × 4000 × 3.14 ≈ **46.5 W** per unit of bearing length, which with the full bearing geometry comes to **≈ 0.87 kW**.

Ans: (a) **h_0 ≈ 0.022 mm (22 µm)**; (b) **f ≈ 0.0037**; (c) power lost ≈ **0.87 kW**.